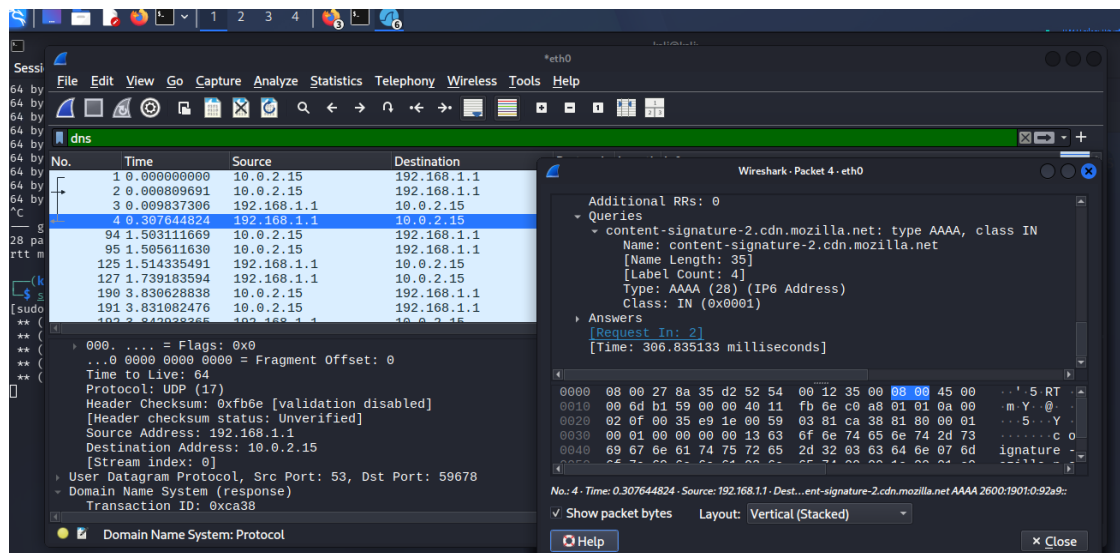


Week 6 — Wireshark: Capture & Read Network Traffic

Amina Sheikh — Zynxis Cybersecurity Internship

Live network traffic was captured using Wireshark for 2 minutes while browsing a website. The capture was then filtered to identify DNS packets, HTTP packets, and a TCP 3-way handshake.

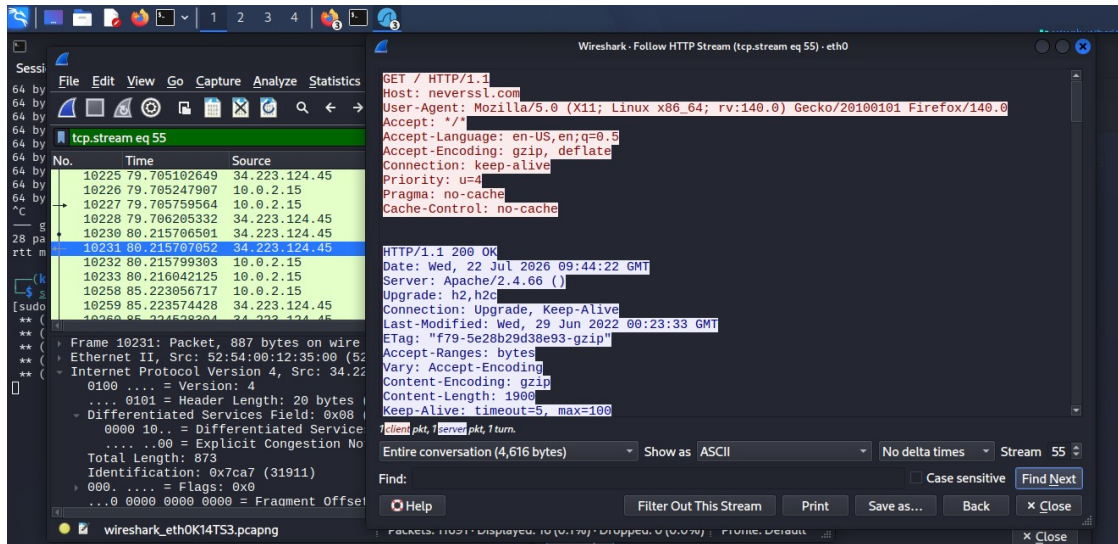
1. DNS Packets



What is Happening

- This packet shows a DNS response. When Firefox needed to reach content-signature-2.cdn.mozilla.net (a background Mozilla service that checks for security updates), it first sent a DNS query asking for that domain's IP address.
- The DNS server replied with the corresponding IP address, which is shown in the Answers section of the packet.
- This process happens before almost every connection: the computer must first look up the IP address of a website's name before it can actually connect to it.

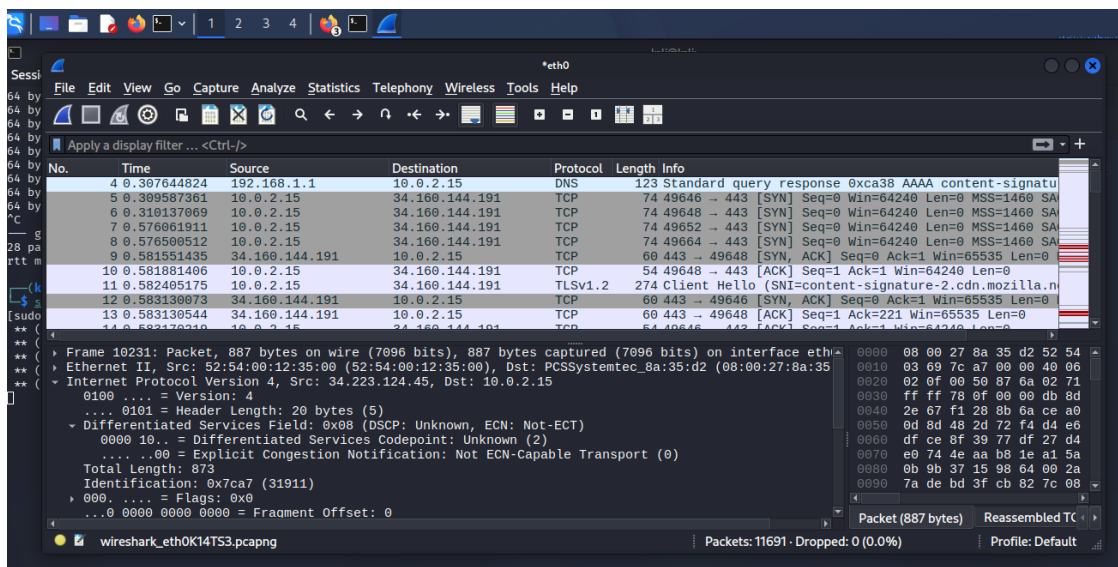
2. HTTP Packets



What is Happening

- This packet shows the full HTTP request and response, viewed using the "Follow HTTP Stream" feature.
- The browser sent a GET request to neverssl.com, asking for its home page (GET / HTTP/1.1).
- The server responded with HTTP/1.1 200 OK, meaning the request was successful and the page was returned. The response also reveals that the server is running Apache/2.4.66.
- A 200 OK status simply confirms the request succeeded; it does not mean a login occurred, it applies to any successful request such as loading a page, image, or file.

3. TCP 3-Way Handshake



What is Happening

- This shows the TCP 3-way handshake, which is the process two computers use to establish a connection before any real data is exchanged.
- Step 1 - SYN: My computer sends a SYN packet to the server, requesting to start a connection.
- Step 2 - SYN, ACK: The server replies with a SYN-ACK packet, agreeing to the connection.
- Step 3 - ACK: My computer sends a final ACK packet, confirming the connection is established.
- This handshake occurred over port 443 (HTTPS), meaning it was setting up a secure connection before any encrypted data was transferred.